PADHAKU
Curiosity · Class 6 Science



CHAPTER 3

Mindful Eating: A Path to a Healthy Body

NCERT Exercise Solutions

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✨ Try each question yourself first, then check the answer. 💡
answers are samples: write your own.

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⭐ 3.1 What Do We Eat?

Activity 3.1 **List the food items you have consumed over the week in Table 3.1.**

What observations can you make about your food? Do you eat the same kind of food in every meal or do your choices vary? Compare your list with your friends' lists. What did you find?

💡 **SAMPLE ANSWER**

Sample table (fill in your own food):

Day	Food items
Monday	Roti, dal, rice, potato sabzi, milk
Tuesday	Poha, rajma-rice, curd, banana
Wednesday	Paratha, khichdi, salad, buttermilk
Thursday	Idli, roti, chana, guava
Friday	Upma, rice, sambhar, egg/paneer
Saturday	Dalia, roti, mixed vegetables, orange
Sunday	Puri-chhole, kheer, rice, dal

Observations: we do not eat the same food in every meal; our choices vary from meal to meal and day to day.

My friends and I eat some common items (roti, rice, dal), but many items are different. There is a **variety** in the food eaten by different people.

? **Do you think that such diversity in food exists in all states of our country?**

✔ ANSWER

Yes. Every state of India has its own traditional food, so there is great diversity in food across the country.

⭐ 3.1.1 Food in different regions

Activity 3.2 **Find out the types of food traditionally consumed and the crops grown in various states of India. Add more states to Table 3.2 and fill in the data.**

💡 SAMPLE ANSWER

Some more states (sample):

State	Locally grown crops	Traditional food items	Beverages
West Bengal	Rice, jute, mustard, fish (from water bodies)	Rice, macher jhol (fish curry), shukto, mishti doi, rasgulla	Tea, milk
Gujarat	Bajra, groundnut, cotton, wheat	Dhokla, thepla, khichdi-kadhi, bajra rotla, undhiyu	Chhaas (buttermilk), tea

Kerala	Rice, coconut, banana, spices	Puttu, appam, avial, sadya, banana chips	Coffee, tea, coconut water
Rajasthan	Bajra, jowar, pulses	Dal-baati-churma, bajra roti, gatte ki sabzi	Chhaas, lassi

? Why do we see diversity in traditional food consumed in various states of our country?

✔ ANSWER

India is a big country with different soil and climate in different regions, so different crops are grown in different states. People also have different taste preferences, cultures and traditions. So the traditional food differs from state to state.

? Analyse the data in Table 3.2. Are there food items that are common across states? Make a list of those food items.

💡 SAMPLE ANSWER

Yes. Some common items are rice, roti (wheat), dal (pulses), curd/buttermilk, milk and tea. Some items are eaten only in one state, e.g. makki di roti (Punjab), ragi mudde (Karnataka), eromba (Manipur).

? **What relation do you find between the traditional food items and the locally grown crops?**

✔ ANSWER

The traditional food of a state is usually made from the crops grown in that state. For example, Punjab grows maize and wheat, so makki di roti and parantha are eaten there; Karnataka grows rice and ragi, so idli, dosa and ragi mudde are eaten there.

⭐ 3.1.2 How have cooking practices changed over time?

Activity 3.3 **Prepare a list of questions to ask elderly people about their food habits and cooking practices, and interview them.**

Sample questions: What kind of food do you still eat and what is new? What are the changes in cooking practices over time? What has caused these changes?

💡 SAMPLE ANSWER

More questions Ravi/you could ask (sample):

- What fuel did you use for cooking earlier?
- What utensils were used (clay, brass, iron) and what do we use now?
- Did you eat more millets earlier? Do you now eat more packaged food?

Sample findings: Earlier people cooked on a chulha with wood or cow-dung cakes, ground spices on a sil-batta, used clay and brass pots and ate more millets and home-made food. Now we use gas stoves, mixer-grinders, pressure cookers and more packaged food.

? What are your findings from the interviews you conducted?

💡 SAMPLE ANSWER

Cooking (culinary) practices have changed a lot. Chulha has been replaced by gas stove, sil-batta by electrical grinder, and cooking is now faster and easier.

? Find out what were the other ways of cooking and grinding.

💡 SAMPLE ANSWER

- Cooking: in a clay oven (tandoor), on angithi (coal stove), in clay pots (handi), roasting on hot ashes or sand.

- Grinding: chakki (hand-operated stone mill) for flour, okhli-musal (mortar and pestle) for pounding grains and spices.

? Why have these culinary practices changed over time?

✔ ANSWER

Because of technological development (new machines like gas stoves and grinders), improved transportation and better communication. Modern methods also save time and effort.

⭐ 3.2 What are the Components of Food?

? Have you ever missed a meal? How do you feel when you miss a meal?

💡 SAMPLE ANSWER

Yes. When I miss a meal I feel tired, weak, hungry and less energetic, and I cannot concentrate well.

? Why do you think a marathon runner drinks glucose water during and after a race?

✔ ANSWER

Running uses a lot of energy. Glucose (a carbohydrate) gives **instant energy**, so the runner does not feel tired.

? Why do you think we prefer to have laddoos as a part of our traditional diet in winters?

✔ ANSWER

Laddoos are made of besan/wheat flour (carbohydrates) with ghee, nuts and seeds (fats). They give a lot of energy and keep our body warm in winter.

? Identify more food items that are rich sources of carbohydrates and fats.

✔ ANSWER

- Carbohydrates: jowar, ragi, bread, chapati, sugar, jaggery (gur), honey, oats, chiku, grapes.
- Fats: ghee, butter, cream, mustard oil, cheese, egg yolk, cashew nuts, sesame (til), mustard seeds.

? Which of these food components are part of your daily diet?

💡 SAMPLE ANSWER

Carbohydrates (rice, roti), proteins (dal, milk, paneer/egg) and fats (ghee, oil) are part of my daily diet.

? **Why do you think we are advised to include servings of fruits, vegetables and other plant-based foods in our daily diet?**

✔ ANSWER

Fruits and vegetables give us vitamins, minerals, fibre (roughage) and water. They protect us from diseases and keep our digestion healthy.

Case 1 **What do you interpret by reading Case 1 (sailors with bleeding and swollen gums recovered after eating lemons and oranges)? What cures scurvy?**

✔ ANSWER

Bleeding and swollen gums are symptoms of **scurvy**, caused by lack of **Vitamin C**. Lemons and oranges (citrus fruits) are rich in Vitamin C, so eating them cures scurvy.

Case 2 **What do you infer from Case 2 (swelling at the front of the neck reduced after using iodised salt)?**

✔ ANSWER

The soil of that region lacked iodine, so local food and water had too little iodine. Lack of iodine causes **goitre** (swelling at the front of the neck). Using iodised salt gives the body enough iodine and prevents goitre.

? **You may have learnt about iodised salt from newspapers, advertisements or a salt packet. What does it mean?**

✔ ANSWER

Iodised salt is common salt mixed with the required quantity of salts of iodine.

? **How would you find out more about other food components that protect our body from various diseases?**

✔ ANSWER

By reading books and charts (like Fig. 3.5), searching the internet, asking a doctor or teacher, and doing a survey of people in the neighbourhood (Activity 3.4).

Activity 3.4 **Study the chart in Fig. 3.5. Find out more sources of vitamins and minerals.**

Survey your neighbourhood: does anyone show the symptoms listed? Correlate them with diet, identify the

deficiency disease, suggest causes, changes in diet and a visit to a doctor.

💡 SAMPLE ANSWER

More sources (sample):

- Vitamin A: spinach, pumpkin, egg yolk
- Vitamin B1: rice bran, peas, sunflower seeds
- Vitamin C: tomato, amla, lemon, sprouts
- Vitamin D: sunlight, egg, fish, mushroom
- Calcium: milk, ragi, sesame seeds
- Iron: spinach, jaggery, dates, pulses
- Iodine: iodised salt, seaweed

Sample survey result: A person feels weak and short of breath and eats few green vegetables → may have **anaemia** (iron deficiency). Advice: eat green leafy vegetables, beetroot, pomegranate, jaggery, and visit a doctor.

? **What differences do you observe in raw and cooked vegetables?**

✔ ANSWER

Cooked vegetables often lose their bright colour and become softer and less crisp. Some nutrients like vitamin C are lost during cooking due to high heat.

? Would it not be wise to include fruits and uncooked vegetables into our diet?

✔ ANSWER

Yes. Fresh fruits and raw vegetables (salads) keep all their vitamins, so we should eat them regularly, after washing them well.

? Do we obtain all nutrients from plants and animals?

✔ ANSWER

No. Some nutrients like salt (a mineral) are obtained from seawater or rocks. Vitamin D can also be made by our body in sunlight.

? What are the food sources that provide water to our body? List a few of them.

✔ ANSWER

Watermelon, cucumber, orange, coconut water, tomato, milk, buttermilk, curd, soups and dal.

⭐ 3.3 How to Test Different Components of Food?

Activity 3.5 **Test for starch: put 2–3 drops of diluted iodine solution on small pieces of potato, cucumber, bread, boiled rice, boiled gram, crushed peanuts, oil,**

butter and coconut. Have they turned blue-black?

✔ ANSWER

Observation: Potato, bread, boiled rice and boiled gram turn **blue-black**. Cucumber, oil, butter and coconut do not; peanuts show little or no change.

Conclusion: A blue-black colour shows the presence of starch. Potato, bread, rice and gram contain starch.

Activity 3.6

Test for fats: wrap each food item in paper and press it (let water dry). Does the paper develop an oily patch? What is the reason? Which items contain fats?

✔ ANSWER

Observation: Peanuts, butter, oil and coconut leave an **oily (translucent) patch**; light shines faintly through it. Potato, cucumber, rice, bread and boiled gram do not.

Reason/Conclusion: The oil or fat in the food makes the patch. Peanuts, butter, oil and coconut contain fats.

Activity 3.7

Test for proteins (teacher demonstration): paste of food + water in a test tube, add 2 drops of copper sulfate solution and 10 drops of caustic

soda solution, shake and wait. Did some test tubes turn violet?

✔ ANSWER

Observation: Tubes with boiled gram, peanuts, soya bean and peas turn **violet**. Potato, cucumber, rice and butter do not (bread may turn slightly violet).

Conclusion: Violet colour shows the presence of proteins. Gram, peanuts, soya bean and peas contain proteins.

Table 3.3 **Record your observations in Table 3.3 (starch, fat, protein present: Yes/No).**

💡 SAMPLE ANSWER

Expected results:

Food item	Starch	Fat	Protein
Potato	Yes	No	No
Cucumber	No	No	No
Boiled rice	Yes	No	No (very little)
Boiled gram	Yes	No	Yes
Peanuts	No (very little)	Yes	Yes
Bread/Chapati	Yes	No	Yes (a little)
Butter	No	Yes	No
Coconut	No	Yes	No (very little)

Your own results may differ slightly.

? **What conclusions can you draw from Table 3.3? Which food items show the presence of more than one nutrient? Which show both proteins and fats?**

✔ ANSWER

Most food items contain more than one nutrient. Boiled gram (starch + protein), bread (starch + protein) and peanuts show more than one nutrient. **Peanuts** show both proteins and fats. So any food we eat may contain many nutrients.

? **Is there a food item that lacks any of these nutrients? Which of these foods do you consume daily?**

💡 SAMPLE ANSWER

Cucumber does not show starch, fat or protein (it is mostly water, with fibre, vitamins and minerals). I eat rice, chapati, gram/dal and butter almost daily.

? **Try to find out other foods that are good sources of starch, fats and proteins.**

✔ ANSWER

- Starch: wheat, maize, sweet potato, bajra, sago (sabudana).
- Fats: ghee, mustard oil, cheese, almonds, walnuts.

- Proteins: milk, paneer, eggs, fish, moong, rajma, soya bean.

⭐ 3.4 Balanced Diet

? **Are nutritional requirements the same for everyone? Do you and your grandparents need the same type or the same amount of nutrients?**

✔ ANSWER

No. The type and amount of nutrients needed depend on age, gender, physical activity, health status and lifestyle. A growing child needs more proteins for growth, while grandparents need less energy food but enough calcium, vitamins and fibre.

Activity 3.8 **Check whether the food you listed in Activity 3.1 contains all the nutrients and other essential components. If not, which need to be added?**

💡 SAMPLE ANSWER

Sample: My diet has enough carbohydrates (rice, roti) and fats (ghee, oil), but few fruits and green leafy vegetables. So vitamins, minerals and roughage should be added, and more protein (dal, milk, paneer, sprouts).

? **What changes would you make in your diet to make it a balanced diet?**

💡 **SAMPLE ANSWER**

- Eat a fruit and a salad every day.
- Add green leafy vegetables and dal/sprouts/milk daily.
- Eat millets (ragi, bajra) sometimes.
- Drink enough water; avoid junk food.

Activity 3.9 **Compare the nutritional information (per 100 g):**

Nutrient	Potato wafers	Roasted chana
Energy	536 kcal	355 kcal
Fats	35.0 g	6.26 g
Carbohydrates	53.0 g	58.58 g
Proteins	7.0 g	18.64 g
Dietary fibre	4.8 g	16.8 g

Which food would you choose? Why?

✔ **ANSWER**

I would choose **roasted chana**.

It has much less fat (6.26 g vs 35.0 g), much more protein (18.64 g vs 7.0 g) and much more dietary fibre (16.8 g vs 4.8 g). Potato wafers give high calories mainly from fat.

? **Which of the two foods you studied in Activity 3.9 could be labelled as junk food?**

✔ ANSWER

Potato wafers, because they are high in fat and calories but low in proteins and dietary fibre.

⭐ 3.5 Millets: Nutrition-rich Cereals

? **Have you ever had food items made from these millets (jowar, bajra, ragi, sanwa)?**

💡 SAMPLE ANSWER

Yes. For example bajra roti, jowar bhakri, ragi dosa or ragi laddoo, and sanwa (samak) khichdi during fasts.

⭐ 3.6 Food Miles: From Farm to Our Plate

? **How would eating local food help reduce food miles?**

✔ ANSWER

Local food is grown near our home, so it travels a short distance from the farm to our plate. This reduces food miles, transport cost and pollution, keeps food fresh and supports local farmers.

? **Try to find the timeline for the various processes involved in getting the food (chapati) from farm to plate (Fig. 3.10).**

💡 SAMPLE ANSWER

Approximate timeline (sample):

- Growing wheat (sowing to harvest): about 4–5 months
- Threshing and winnowing: a few days
- Storage of grains: a few weeks to months
- Grinding and packing: 1–2 days
- Transport to retail shop: a few days
- Cooking chapati at home: a few minutes

So it takes many months of effort by farmers and others. We must not waste food.

🔭 Let us enhance our learning

1 (i) **Pick the odd one out and give reasons: Jowar, Bajra, Ragi, Chana**

✔ ANSWER

Chana is the odd one out. Jowar, bajra and ragi are millets (cereals, rich in carbohydrates); chana is a pulse (rich in protein).

1 (ii) **Pick the odd one out and give reasons: Kidney beans, Green gram, Soya bean, Rice**

✔ ANSWER

Rice is the odd one out. It is a cereal, rich in carbohydrates; the others are pulses/legumes, rich in protein.

2 **Discuss traditional versus modern culinary practices in India.**

💡 **SAMPLE ANSWER**

Traditional	Modern
Cooking on chulha with wood/cow-dung cakes	Cooking on gas stove, induction, microwave
Grinding on sil-batta, chakki	Electrical mixer-grinder, packaged flour
Clay, brass, iron utensils	Steel, aluminium, non-stick cookware, pressure cooker
Slow, more effort, fresh home-made food	Fast and easy, more packaged/processed food

Modern methods save time and effort; traditional methods use more local, fresh and less processed food, which is healthy.

3 **A teacher says that good food may act as medicine. Ravi is curious and has some questions. List at least two questions that he can ask.**

💡 **SAMPLE ANSWER**

- Which foods can help us recover from or prevent diseases?

- How do lemons and oranges cure scurvy?
- Which food should I eat to avoid anaemia or night blindness?
- Can eating good food reduce the need for medicines?

4 **Not all delicious foods are necessarily healthy, while not all nutritious foods are always enjoyable. Share your thoughts with a few examples.**

💡 **SAMPLE ANSWER**

Many tasty foods like potato wafers, noodles, pizza, candy bars and carbonated drinks have high sugar and fat but few nutrients, so they are not healthy if eaten often.

Some nutritious foods like bitter gourd (karela), spinach, sprouts and ragi may not taste good to everyone, but they are rich in vitamins, minerals and fibre.

We should choose healthy food and cook it in tasty ways, e.g. spinach paratha, ragi laddoo, sprouts chaat.

5 **Medu does not eat vegetables but enjoys biscuits, noodles and white bread. He often has stomach ache and constipation. What changes should he make in his diet? Explain.**

✔ **ANSWER**

His food lacks **roughage (dietary fibre)**, which helps remove undigested food and ensures smooth passage of stools.

He should eat green leafy vegetables, fresh fruits, salads, wholegrains, pulses and nuts, drink enough water, and reduce biscuits, noodles and white bread.

6 **Reshma had trouble seeing things in dim light. The doctor prescribed a vitamin supplement and advised her to include a few food items in her diet.**

(i) Which deficiency disease is she suffering from?

(ii) Which food component may be lacking in her diet?

(iii) Suggest some food items she should include (any four).

✔ ANSWER

(i) Loss of vision in dim light: **night blindness**.

(ii) **Vitamin A**.

(iii) Papaya, carrot, mango, milk (also spinach, pumpkin).

7 **You are provided: (i) Canned fruit juice (ii) Fresh fruit juice (iii) Fresh fruit. Which one would you prefer and why?**

✔ ANSWER

(iii) Fresh fruit. It has all its vitamins, minerals and dietary fibre (roughage). Fresh juice loses most of the fibre, and canned juice often has added sugar and preservatives and fewer vitamins.

8 **Gourav got a fracture in his leg. The doctor gave him calcium tablets; on the second visit he gave Vitamin D syrup along with calcium tablets. Refer to Fig. 3.5.**

(i) Why did the doctor give calcium tablets?

(ii) Why Vitamin D syrup along with calcium tablets?

(iii) What question arises in your mind about the doctor's choices?

✔ ANSWER

(i) Calcium keeps bones and teeth healthy and strong. It helps the broken bone heal.

(ii) Vitamin D helps the body **absorb calcium** for bone health. Without it, the calcium would not be used well.

(iii) Sample: Why didn't the doctor give Vitamin D on the first visit? Can Gourav get enough calcium and Vitamin D from milk, curd and sunlight instead of medicines?

9 **Sugar is an example of carbohydrates. Sugar is tested with iodine solution but it does not change to blue-black colour. What can be a possible reason?**

✔ ANSWER

The iodine test detects only **starch**. Sugar is a carbohydrate, but it is not starch, so it does not turn blue-black.

10 **What do you think of Raman's statement, "All starches are carbohydrates but not all carbohydrates are starches"? Describe the design of an activity to test your answer.**

✔ ANSWER

The statement is **correct**. Starch is one type of carbohydrate; sugar and glucose are also carbohydrates but are not starch.

Activity: Take boiled rice, a potato slice, and a little sugar and glucose (dissolved in water) in separate dishes. Add 2–3 drops of diluted iodine solution to each.

Result: Rice and potato turn blue-black (starch); sugar and glucose do not. So all carbohydrates are not starches.

11 **Drops of iodine fell on Mishti's socks and on her teacher's saree. The drops on the saree turned blue-black while the socks did not change. What can be a possible reason?**

✔ ANSWER

The saree was probably **starched** (cotton clothes are stiffened with starch), so iodine turned it blue-black. The socks had no starch, so the colour did not change.

12 **Why are millets considered a healthy choice of food? Can eating just millets suffice for the nutritional requirements of the body? Discuss.**

✔ ANSWER

Millets are good sources of vitamins, minerals like iron and calcium, and dietary fibre, so they are called nutri-cereals.

But eating only millets is **not enough**. We also need enough proteins (pulses, milk), fats, vitamins from fruits and vegetables, and water. A balanced diet with a variety of foods is needed.

13 **You are given a sample of a solution. How would you check the possibility of it being an iodine solution?**

✔ ANSWER

Put 2–3 drops of the solution on a food that contains starch, such as a potato slice or boiled rice. If it turns **blue-black**, the solution may be iodine solution.

📘 Learning further

1 **Read the nutritional information of at least three fortified food items and analyse it.**

💡 SAMPLE ANSWER

Sample: Iodised salt (iodine added), fortified atta (iron, folic acid added), fortified milk (Vitamins A and D added). The label shows how much of each nutrient is in 100 g. Fortification adds nutrients that our diet may lack.

2 **The Apatani tribe of Arunachal Pradesh produces a salt called tapyo. Collect information about their salt making process and the need to make their own salt; make a chart and write a paragraph.**

💡 SAMPLE ANSWER

Sample: The Apatani people burn certain plants and collect the ash. Water is passed through the ash, and the liquid collected is dried/boiled to get a salt called tapyo. It is made because their region is far from the sea and salt was hard to get; it adds minerals and flavour to their food.

3 **Read about ranbhajis from Maharashtra and edible mushrooms from Himachal Pradesh. Are you aware of any wild varieties of food from your region? Discuss in class.**

💡 **SAMPLE ANSWER**

Ranbhajis are wild leafy vegetables collected from forests and fields in the monsoon in Maharashtra. Wild edible mushrooms (e.g. guchhi) are collected in Himachal Pradesh.

Examples from other regions (sample): bathua, wild amla, jamun, karonda, mahua flowers, bamboo shoots.

4 **List junk foods you eat frequently; ask friends too. Write a letter to your principal requesting a ban on certain junk foods in school and suggest healthy options.**

💡 **SAMPLE ANSWER**

Sample letter:

To the Principal, ____ School

Subject: Request to ban junk food in the school campus

Respected Sir/Madam,

We surveyed our classmates and found that many of us often eat chips, noodles, candy bars and cold drinks from the school canteen. These foods are high in sugar and fat and low in nutrients. They can cause obesity and other health problems. We request you to ban them in the school campus and provide healthy options like fruits, sprouts chaat, roasted chana, poha, idli and buttermilk.

Yours obediently,

Students of Class 6

5 **Find out the variation in nutritional requirements of different individuals based on age, physical activity and health conditions. Record, discuss and analyse.**

💡 **SAMPLE ANSWER**

Sample findings:

- Growing children need more proteins and calcium.
- Sportspersons and labourers need more energy foods and proteins.
- Elderly people need less energy food but more calcium, vitamins and fibre.
- Pregnant women need more iron, calcium and proteins.

- A person with anaemia needs iron-rich food.

6 **Prepare a diet chart to provide a balanced diet to a twelve-year-old child, with food items that are not expensive and commonly available.**

💡 **SAMPLE ANSWER**

Meal	Food items
Breakfast	Milk, poha/dalia with vegetables, a banana
Mid-morning	Roasted chana or a guava
Lunch	Roti/rice, dal, green leafy sabzi, salad, curd
Evening	Sprouts chaat or ragi laddoo, lemon water
Dinner	Bajra/wheat roti, mixed vegetables, dal or egg/paneer

Drink 6–8 glasses of water daily.

🎉 **Chapter 3 done! Revise the answers once more tomorrow.**